

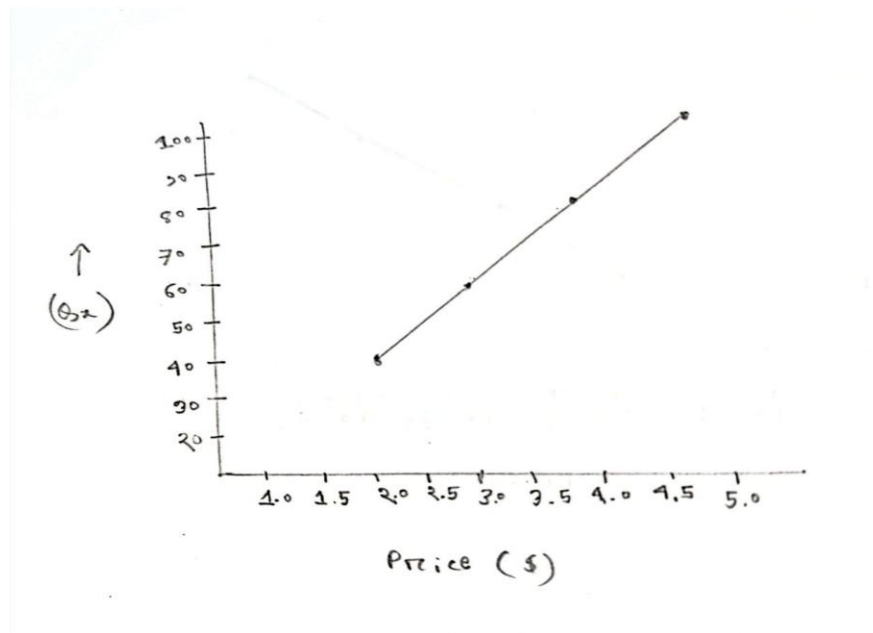
Answer to the Question no. 1

A. Construction of Producer's Supply Schedule

Price(P_x)	Calculation ($20 \times P_x$)	Quantity Supplied (Q_{S_x})
1	20×0	0
2	20×1	20
3	20×2	40
4	20×3	60
5	20×4	80
6	20×5	100

B. Graphical Representation of Supply

The supply curve illustrates the functional relationship between price and the quantity supplied. As shown in the graph, it appears as an upward-sloping linear function originating from zero. This positive slope signifies that an increase in price directly results in a larger quantity supplied.



C. Determinants of Supply & Curve Shifts

The equation $QS_x = 20P_x$ relies on the *ceteris paribus* assumption, meaning it predicts quantity based *only* on price while holding all other variables constant. If external factors change, the entire supply curve shifts rather than moving along the line.

Input Prices: Costs for labor, energy, or raw materials. Higher costs make production expensive, reducing supply (shifting the curve left). Lower costs increase supply (shifting the curve right).

Technology: Improvements in efficiency lower production costs per unit, allowing producers to supply more at the same price (shift right). Technological failures or setbacks reduce supply (shift left).

Taxes & Subsidies: Taxes act as an added cost, decreasing supply (shift left). Subsidies act as financial support, effectively lowering costs and increasing supply (shift right).

Number of Sellers: An increase in the number of firms entering the market boosts total supply (shift right), while firms leaving the market reduces it (shift left).

Future Expectations: If producers anticipate higher prices in the future, they may withhold stock now (shift left). Conversely, expecting lower prices later may prompt them to sell more immediately (shift right).

Prices of Related Goods: If producing alternative goods becomes more profitable, resources will move away from product X (shift left). If alternatives become less attractive, resources move back to X (shift right).

D. Price Threshold Analysis

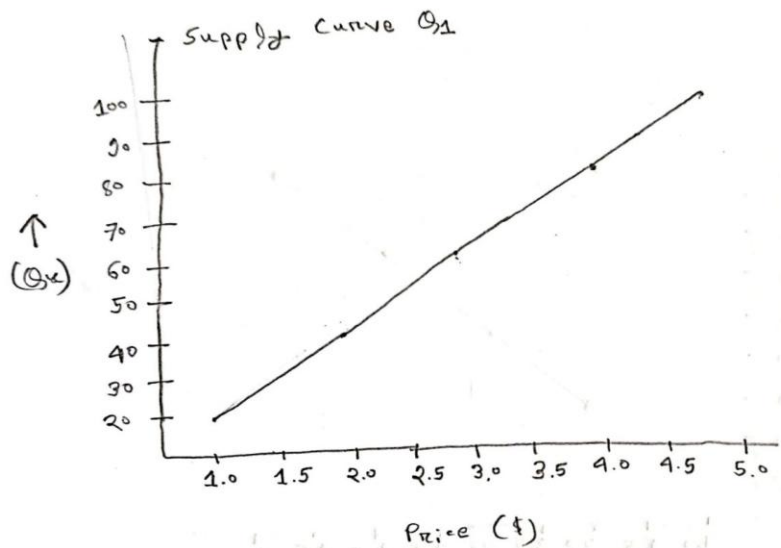
Analyzing the function $QS_x = 20P_x$, we see that quantity supplied is positive for any price greater than zero ($P_x > 0$).

- **Minimum Price:** Since the function implies no fixed costs or specific shutdown price, the producer is willing to supply at any value just above \$0.
- **Unit Calculation:** To supply at least a single unit (1), we solve $1 = 20P$, which results in a price of **\$0.05**. Therefore, for any positive supply to exist, the price must exceed zero.

Answer to the Question no. 2

A. Supply Schedule Dynamics

The graph illustrates two distinct supply scenarios. The green line represents the **initial supply schedule** under normal conditions. The red dashed line represents the **shifted supply schedule**, reflecting a decrease in supply caused by changes in input prices.



B. Analysis of Price Movement (Pre-Shift)

Focusing on the initial curve before any external changes:

- An increase in the price of Commodity X from \$3 to \$5 triggers a movement along the curve.
- According to the initial schedule, this price hike causes the quantity supplied to rise from **30 units** (at \$3) to **40 units** (at \$5). This aligns with the law of supply: higher prices incentivize greater production.

C. Analysis of the Supply Shift

This section isolates the effect of rising input costs (like raw materials):

- At a price of **\$3**, the producer originally supplied 30 units.
- Due to increased production costs, the supply curve shifts inward. On this new schedule, at the same **\$3** price, the producer now only supplies **10 units**.
- **Conclusion:** The rise in input prices forces a reduction in supply from 30 to 10 units at this price point.

D. Net Effect of Simultaneous Changes

This section determines the outcome when both events occur at once: the price rises (\$3 to \$5) *and* input costs increase.

1. **Opposing Forces:** The price increase naturally encourages more supply, but the higher input costs discourage it (shift left).
2. **Quantitative Outcome:**
 - Originally, at $P_x=3$, supply was 30 units.
 - With the price rising to $P_x=5$, the producer *would have* supplied 40 units.
 - However, due to the cost-driven shift, the supply at $P_x=5$ is actually capped at **20 units**.
3. **Final Result:** The negative impact of high input costs outweighs the incentive of the higher price. Consequently, the final quantity supplied (20 units) is lower than the starting quantity (30 units), despite the market price being higher.

Answer to the Question no. 3

A. Elasticity Calculation (Base A)

Calculating the price elasticity of demand moving from Point A to Point B, using Point A as the base:

- Quantity Change (ΔQ): $(41230-3218) = 9,050$
% change in quantity = $(9050/32180) \times 100 = 28.12\%$
- Price Change (ΔP): $(5.70-6.10) = -0.40$
% change in price = $(-0.40/6.10) \times 100 = -6.56\%$

- **Result:**

Elasticity: $(28.12\% / -6.56\%) = -4.29\%$

The elasticity coefficient is **-4.29**. Since the absolute value $|E_d| > 1$, the demand is considered **elastic**, indicating a price decrease leads to a proportionally larger increase in quantity demanded.

B. Elasticity Calculation (Base B)

Calculating the elasticity moving from Point B back to Point A, using Point B as the base:

- **Quantity Change (ΔQ):** $(32180-41230) = -9050$
% change in quantity = $(-9050/41230) \times 100 = -21.95\%$
- **Price Change (ΔP):** $(6.10-5.70) = 0.40$
% change in price = $(0.40/5.70) \times 100 = 7.02\%$

- **Result:**

Elasticity: $(-21.95\% / 7.02\%) = -3.13\%$

The elasticity coefficient is **-3.13**

C. Midpoint (Arc) Elasticity

Using the midpoint formula to find the average elasticity between the two points:

Percentage change in quantity (midpoint)

$$\% \Delta Q = \frac{Q_B - Q_A}{(Q_A + Q_B)/2} = \frac{41230 - 32180}{(32180 + 41230)/2} = \frac{9050}{36705} = 0.2466 \text{ (24.66\%)}$$

Percentage change in price (midpoint)

$$\Delta Q = \frac{P_B - P_A}{(P_A + P_B)/2} = \frac{5.70 - 6.10}{(6.10 + 5.70)/2} = \frac{-0.40}{5.90} = -0.06780 \text{ (-6.78\%)}$$

Elasticity (Midpoint formula)

$$E_P = \frac{\% \Delta Q}{\% \Delta P} = \frac{0.2466}{-0.06780} = -3.64$$

The arc elasticity is roughly -3.64. This confirms the demand is highly elastic in this range.

Graphical Interpretation (Segment A-B)

The segment connecting Point A (6.10, 32,180) and Point B (5.70, 41,230) is downward-sloping, reflecting the inverse relationship between price and quantity. The slope is relatively flat, which visually corresponds to the high elasticity calculated ($|E_d| > 1$). This high sensitivity suggests that consumers react strongly to price changes, implying that **Commodity X likely has close substitutes or is considered a non-essential item** at this price level.

